ALGEBRA 2 · PROBABILITY & STATISTICS

Probability Simulation Lab

Explore & Reflect

Name _____ Date _____ Period _____

Open **Probability Simulation Lab** on Mr. K's Classroom: Algebra 2 → Probability & Statistics → Interactive Math. Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3, then do Part 4 on your own.

1 Predict before you touch the tool

1. If you flip a coin 10 times, will you always get exactly 5 heads?

2. Which sum of two dice do you think comes up most often?

2 Explore use the tool and record what you see

Coin flips

Keep adding flips and record the share of heads.

Total flips	Heads	Experimental P(heads)	Difference from 0.5
10			
100			
1,000			
3,000			

Sum of two dice: theoretical probability

Compare with your experimental bars after 500+ rolls.

Sum	2	4	6	7	9	12
Ways (out of 36)						
Probability						

Normal curve: μ = 70, σ = 8

Shade within	Interval	Percent
1σ		
2σ		
3σ		

3 Write the rule fill in each blank from what you observed

1. Experimental probability = _____. Theoretical probability = _____.

2. Law of large numbers: as the number of trials grows, the experimental probability gets _____ to the theoretical probability.

3. Empirical rule: about _____% of data lies within 1σ, _____% within 2σ and _____% within 3σ.

4. Increasing σ makes the normal curve _____ and _____.

4 Check no tool: show your work

1. What is P(sum = 8) for two dice?

2. Scores have μ = 80 and σ = 5. What percent fall between 75 and 85?

3. Using the same scores, what percent are above 90?

4. You roll one die 600 times. About how many 4s do you expect?

5 Reflect

Why did 10 flips sometimes give a share far from 0.5 while 3,000 flips did not?